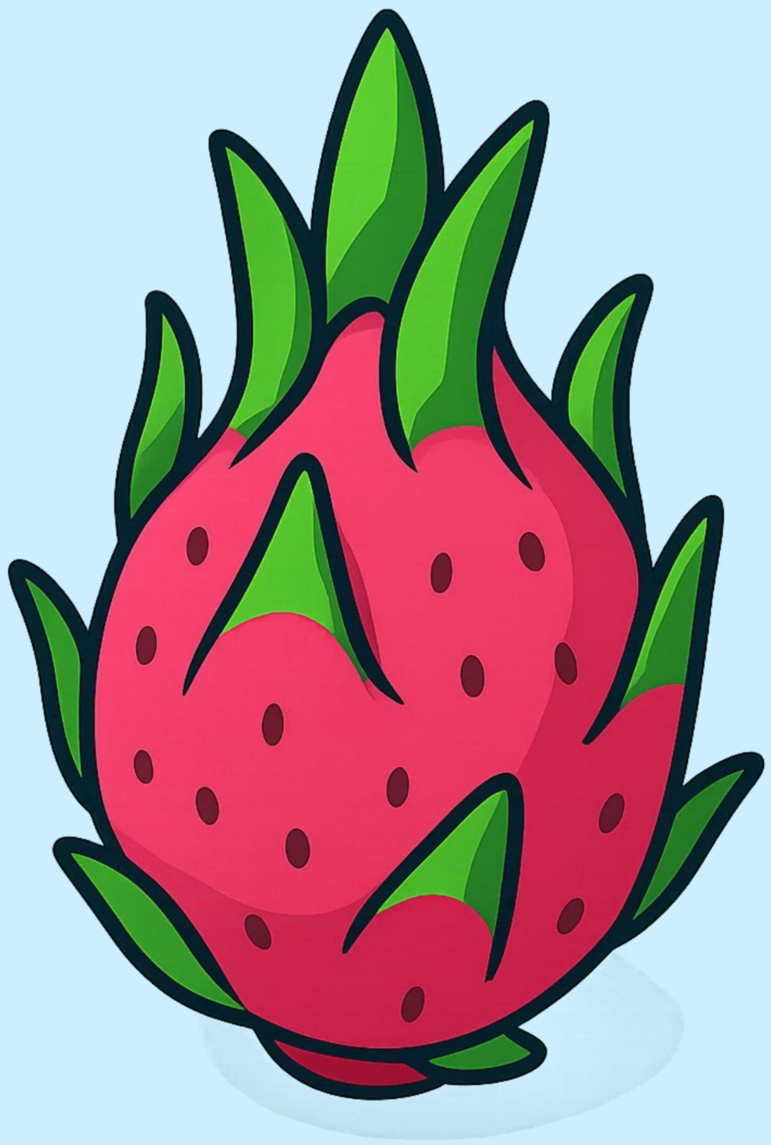
2CEAI603: DEEP NEURAL NETWORK

Deep Learning-Based Classification of Dragon Fruit Leaf and Fruit Condition

This study develops and compares five deep learning architectures — Custom CNN, VGG16, MobileNetV2, ResNet50, and EfficientNetB0 — to automatically classify dragon fruit images into four quality categories: **Good Leaf, Bad Leaf, Good Fruit, and Bad Fruit.**

Author: Aman Khokhar | **Date:** April 12, 2026



The Challenge: Why Automation Matters

Dragon fruit (*Hylocereus spp.*) is a commercially important tropical crop whose leaf and fruit quality directly determines yield and market value. Manual inspection is slow, subjective, and error-prone — creating a clear need for automated, reliable classification systems.

Leaf Diseases

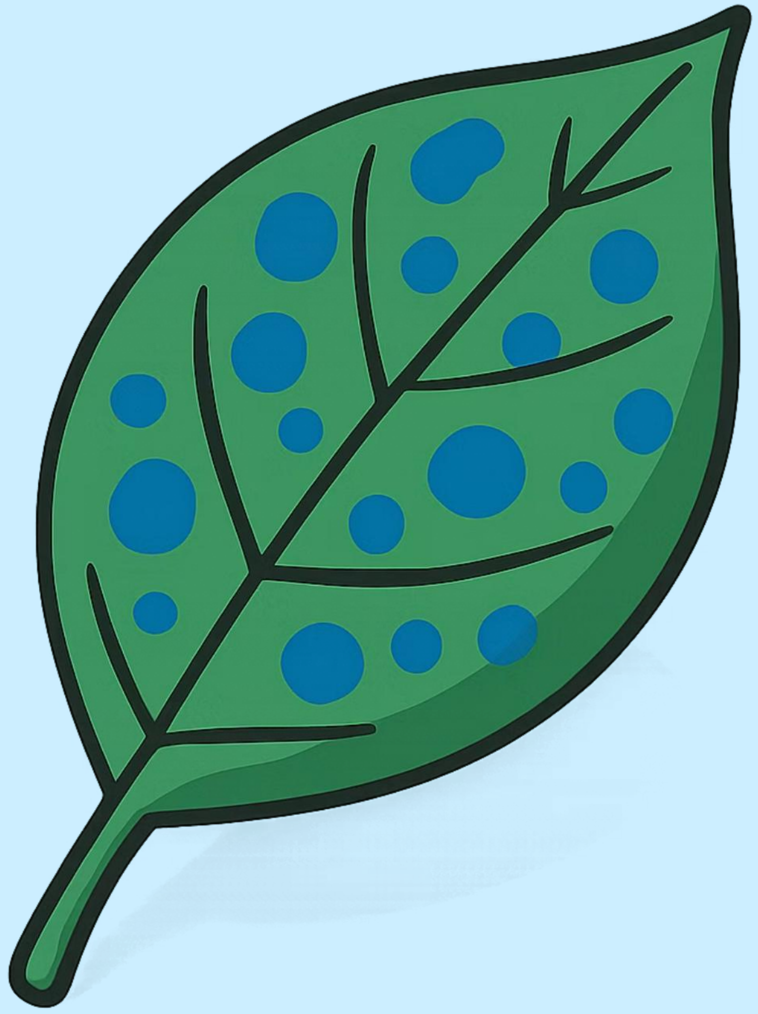
Anthracnose, stem canker, and fungal infections reduce plant productivity and can spread rapidly if undetected.

Fruit Defects

Surface blemishes, discoloration, and rot significantly reduce marketability and export potential.

Four-Class Target

Good Leaf, Bad Leaf, Good Fruit, Bad Fruit — a multi-class problem more complex than binary classification.



Project Scope & Problem Definition

Core Objective

Design and evaluate deep learning models for accurate, automated classification of dragon fruit leaf and fruit conditions from images — reducing human effort and improving grading consistency.

Key Challenges

- Limited dataset size increases overfitting risk
- Class imbalance across the four categories
- Variability in lighting, background, and capture angles
- Multi-class complexity beyond binary classification

Domain & Approach

This work sits at the intersection of **Computer Vision**, **Deep Learning**, and **Agricultural AI**. CNNs automatically learn discriminative features — color gradients, surface texture, leaf venation, and fruit morphology — directly from raw pixels, eliminating manual feature engineering.

Pre-trained models leverage ImageNet weights via transfer learning, significantly boosting performance even with limited domain-specific data.

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Five Architectures Under Evaluation

A deliberate spectrum of model complexity — from a shallow custom CNN to deep, pre-trained backbones — enables a comprehensive comparison of accuracy, efficiency, and suitability for small agricultural datasets.

Custom CNN

Built from scratch. No pre-trained weights — fully dependent on the available dataset for feature learning.

VGG16

16 weight layers using 3×3 filters throughout. A classical, well-established baseline for transfer learning in plant disease detection.

MobileNetV2

Lightweight inverted-residual architecture. Optimized for speed and efficiency — ideal for real-time and resource-constrained deployment.

ResNet50

50-layer residual network with skip connections. Powerful but data-hungry — requires careful fine-tuning on smaller datasets.

EfficientNetB0

Compound-scaled architecture balancing depth, width, and resolution. Efficient by design but sensitive to dataset size and hyperparameters.

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Solution Methodology: End-to-End Pipeline

Collect & Prep

Augmentation

Train Models

Evaluate

Select & Deploy

All images are resized to **224 × 224 pixels** and normalized before training. Augmentation techniques — random rotation, horizontal flipping, and zoom — artificially expand the dataset and improve generalization. Models train for up to 100 epochs with early stopping (patience = 10) to prevent overfitting. For transfer learning models, base layers are initially frozen; selective fine-tuning is then applied to the final convolutional blocks. The final classification layer uses **Softmax activation** with categorical crossentropy loss and the Adam optimizer.

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Handling Data Limitations

Data Augmentation Strategy

With a relatively small dataset, augmentation is critical to prevent overfitting and improve generalization. The following transformations are applied on-the-fly during training:

- **Random rotation** — simulates varied capture angles
- **Horizontal & vertical flipping** — increases orientation diversity
- **Random zoom** — mimics different camera distances

These techniques collectively increase effective dataset size and expose models to a broader range of visual variations.

Class Imbalance & Learning Rate Tuning

Class weights are computed inversely proportional to class frequencies, ensuring underrepresented categories contribute equally to the loss function during training.

Three learning rates were systematically evaluated across all five models:

LR = 0.1

Unstable — overshoots loss minima

LR = 0.01

Moderate — acceptable convergence

LR = 0.001

Optimal — stable, highest accuracy

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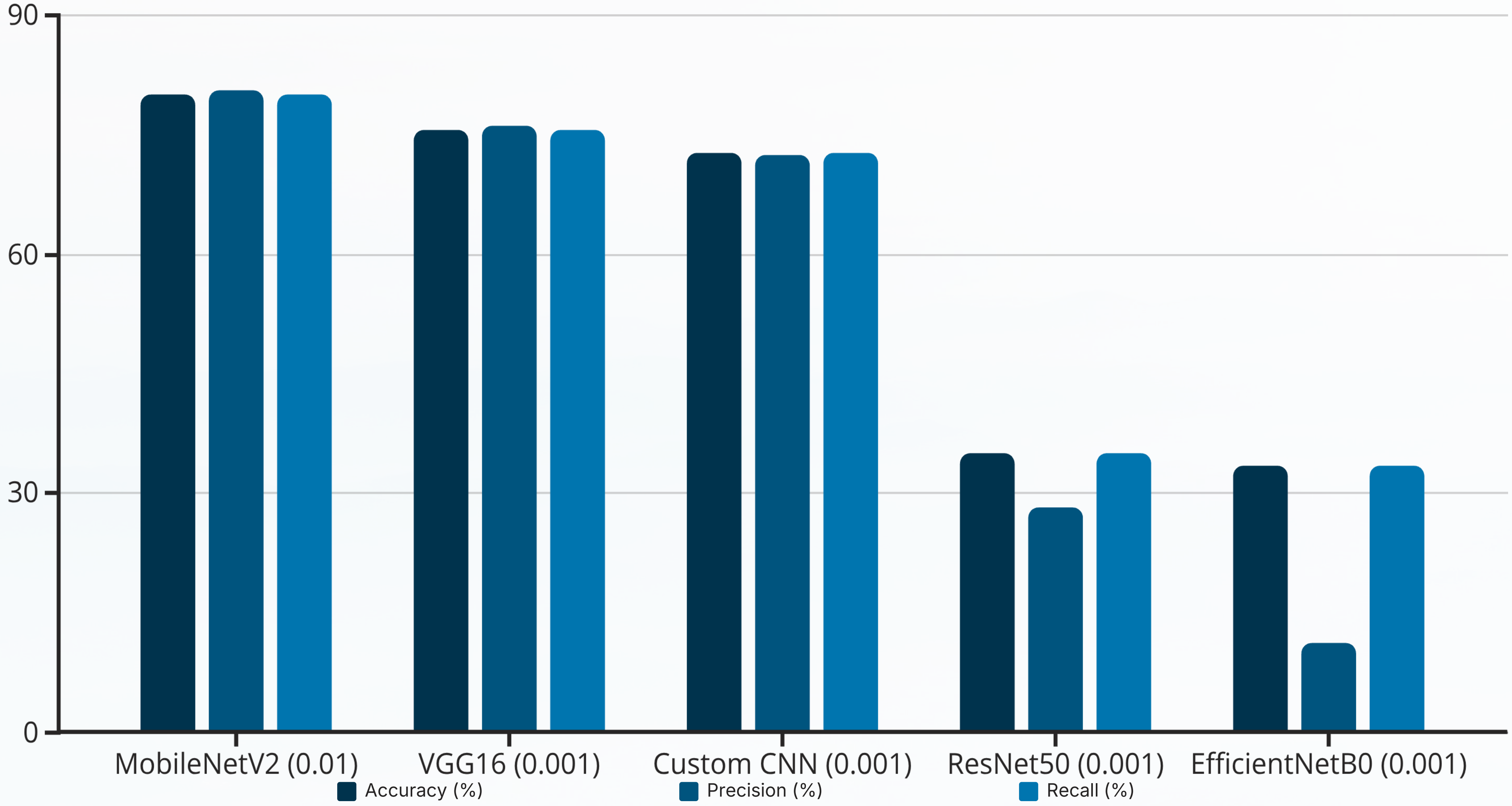
Results: Model Performance at a Glance

Performance metrics are reported for each model at its best-performing learning rate. **MobileNetV2** achieved the highest accuracy (80.0% at LR=0.01), while **VGG16** delivered strong, consistent results (75.6% at LR=0.001). ResNet50 and EfficientNetB0 struggled due to their complexity relative to the available dataset size.



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Comparative Analysis & Justification

Model	Best LR	Accuracy	Precision	Recall	AUC
MobileNetV2	0.01	80.00%	80.64%	80.00%	0.850
VGG16	0.001	75.56%	76.22%	75.56%	0.824
Custom CNN	0.001	72.78%	72.58%	72.78%	0.796
ResNet50	0.001	35.00%	28.10%	35.00%	0.513
EfficientNetB0	0.001	33.33%	11.11%	33.33%	0.500

Why MobileNetV2 Won

Its lightweight inverted-residual design generalizes efficiently on small datasets without overfitting. Pre-trained ImageNet features transfer well to the four-class dragon fruit problem.

Why ResNet50 & EfficientNetB0 Underperformed

Both architectures are highly complex and require large datasets and extensive fine-tuning. With limited data, they failed to converge meaningfully — accuracy hovered near random chance (33%).

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Conclusions & Key Takeaways

Deep Learning Works for Agricultural Quality Grading

All five CNN architectures successfully classified dragon fruit conditions into four categories, validating the feasibility of automated quality detection in real-world agricultural settings.

MobileNetV2 is the Recommended Architecture

With 80% accuracy and a lightweight footprint, MobileNetV2 is best suited for deployment — particularly in resource-constrained or edge environments such as mobile field applications.

Data Quality & Augmentation Are Critical

Augmentation, class weighting, and careful learning rate selection (0.001) were essential to achieving competitive results. LR = 0.1 consistently caused training instability across all models.

VGG16 Remains a Strong Baseline

Despite its age, VGG16's deep sequential 3×3 filter architecture delivered 75.56% accuracy — confirming its continued relevance as a reliable transfer learning backbone for plant image classification.

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Future Directions & Roadmap

Several promising avenues exist to extend this work from a research prototype toward a production-ready agricultural tool.

1 Expand the Dataset

Collect a larger, more diverse dataset across multiple farms, seasons, and lighting conditions to improve model robustness and generalization.

2 Grad-CAM Explainability

Apply Gradient-weighted Class Activation Mapping to visualize which image regions drive predictions — building trust with farmers and quality inspectors.

3 Ensemble & Fine-Tuning

Explore ensemble methods combining MobileNetV2 and VGG16 predictions, and apply full fine-tuning of transfer learning models with expanded data.

4 Object Detection (YOLO)

Transition from image classification to object detection frameworks (e.g., YOLO) to simultaneously locate and classify multiple defects within a single image.

5 Mobile & Edge Deployment

Deploy the best-performing model as a lightweight mobile app or edge device application, enabling real-time field-level quality grading for farmers and exporters.

Acknowledgements: Thanks to the faculty for guidance, the institution for resources, and the open-source community — TensorFlow, Keras, and the Kaggle dragon fruit dataset — for supporting this work.

Aman Khokhar